

CCNA 200-301 - Chapter 1

Network Fundamentals (Part 1)

Key Notes

- Network: A group of connected devices used to exchange data and share resources.
- Main benefits: File sharing, Internet access, Printers, Communication.

Network Types

- PAN: Personal devices (Bluetooth, smartwatch).
- LAN: Home, office, school. High speed.
- MAN: Connects multiple LANs inside one city.
- WAN: Connects cities/countries. Internet is the biggest WAN.

Network Topologies

- Star: Most common. All devices connect to a switch.
- Bus: Single backbone cable; cable failure affects the network.
- Ring: Devices connected in a loop.
- Mesh: Highest redundancy, highest cost.

Network Devices

- Switch → Connects devices in the same LAN (Layer 2, MAC Address).
- Router → Connects different networks (Layer 3, IP Address).
- Access Point → Provides Wi-Fi access.
- Modem → Connects to the ISP.
- Server → Provides services (Web, File, DNS, DHCP).

Transmission Media

- Wired: UTP, STP, Fiber Optic.
- Wireless: Wi-Fi, Bluetooth.

Must Remember

- PAN < LAN < MAN < WAN.
- Switch = Layer 2 = MAC Address.

- Router = Layer 3 = IP Address.
- Star topology is the most common in modern networks.

Commands

No Cisco IOS commands in this lesson.